

# End-to-End Test Scenario – Inbound and Outbound Operations

This test scenario validates the full life cycle of inventory through the Labamu Warehouse Management System (WMS) from purchase order creation through receiving, putaway, sales order processing, picking and shipping. It mimics the daily operations of a small warehouse and ensures that inventory quantities, statuses and routes update correctly throughout each stage. The scenario assumes the updated WMS with one-to-many area-to-location mapping and configurable origin is in place.

## Test Objectives

- Verify that purchase orders can be created, confirmed and received, updating inventory quantities and purchase order status.
- Ensure inbound receipts generate putaway tasks that assign stock to the correct locations according to putaway rules or manual selection.
- Confirm that sales orders can be created and processed through reservation, picking, packing and shipping, decrementing on-hand inventory and updating order status.
- Validate that the dashboard metrics (stock value, fulfilment rate, pending orders and warehouse capacity) reflect the transactions performed.

## Pre-conditions and Setup

Before executing the end-to-end scenario, set up the following data and configuration:

1. **Warehouse and Locations**
2. Confirm a warehouse exists (e.g., *Main Warehouse*) with an associated floor plan. Ensure the functional areas (e.g., *Receiving Dock*, *Main Storage*, *Shipping Dock*) have been linked to their respective locations (zones and bins). For example, the *Main Storage* area may map to multiple bin locations such as *MS-A1*, *MS-A2*, etc.
3. Create or confirm the existence of location codes within these areas. Each location should include dimensions and attributes appropriate for the products.
4. **Product Master Data**
5. Create a product *Pro Laptop X* (SKU: LAP-X) with category *Electronics*. Set unit of measure, ABC/velocity class and packaging details as necessary.
6. Ensure the product has a defined cost and selling price.
7. **Supplier and Customer**

8. Create a supplier record (e.g., *PT TechSupplier*). Enter contact, payment terms and shipping address.
9. Create a customer record (e.g., *CV Gadget Store*) with billing and shipping details.

## 10. Putaway Rules

11. Define at least one putaway rule for *Pro Laptop X* or its category. For example, match by category *Electronics* with a destination strategy of *Main Storage* area sorted by *least occupied* location. This ensures that receiving tasks are directed to the correct bins.

## 12. Permissions

13. Ensure that test users have the necessary roles (e.g., *Warehouse Manager*, *Inventory Clerk*, *Picker*) and permissions to create purchase orders, receive goods, process putaway, create sales orders and perform picking and shipping.

## End-to-End Steps

### 1. Purchasing (Create and Confirm Purchase Order)

| Step | Action                                                                                                                                                                                                                                                         | Expected Result                                                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1.1  | In the <b>Suppliers &gt; Purchase Orders</b> module, click <b>New Purchase Order</b> . Select <i>PT TechSupplier</i> as the vendor, choose the <i>Main Warehouse</i> and add the line item <i>Pro Laptop X</i> with a quantity of 20 and the agreed unit cost. | A draft purchase order is created showing the supplier, warehouse, item, quantity and total cost.                                       |
| 1.2  | Save and <b>Confirm</b> the purchase order.                                                                                                                                                                                                                    | The purchase order status changes to <i>Confirmed</i> (or <i>Ordered</i> ). The system generates an expected arrival date and quantity. |

### 2. Receiving (Inbound Delivery)

| Step | Action                                                                                                                                                                                                                                      | Expected Result                                                                                                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1  | Navigate to <b>Inbound &gt; Putaway</b> or <b>Inbound &gt; Receiving</b> (depending on UI). Locate the recently confirmed PO and click <b>Receive</b> . Enter the quantity received (20) and record any lot numbers or serials if required. | A receipt record is created, and the purchase order line shows 20 units received. The PO status updates to <i>Received</i> or <i>Done</i> . Inventory on hand increases by 20 units in the <i>Receiving Dock</i> staging area. |
| 2.2  | Check the <b>Lot Number</b> module. If lot tracking is enabled, ensure that a lot record has been created with the correct product, quantity and expiration dates.                                                                          | A new lot number record exists for <i>Pro Laptop X</i> with 20 units and is associated with the receiving location.                                                                                                            |

### 3. Putaway

| Step | Action                                                                                                                                                             | Expected Result                                                                                                                                                                                                                                           |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1  | In the <b>Inbound &gt; Putaway</b> module, start a new putaway session for the <i>Main Warehouse</i> . The system displays tasks for the received goods.           | One or more putaway tasks appear, each showing the source location ( <i>Receiving Dock</i> ) and recommended destination location(s) based on the putaway rule (e.g., <i>MS-A1</i> ).                                                                     |
| 3.2  | Accept the tasks and direct the operator to physically move the goods to the suggested bins. Use a mobile scanner (if available) to scan the destination location. | As each task is completed and validated, the system moves inventory from the <i>Receiving Dock</i> to the destination bins. The putaway session shows the task status as <i>Done</i> or <i>Closed</i> . On-hand quantity now resides in the storage bins. |

### 4. Sales and Order Management

| Step | Action                                                                                                                                                                                                    | Expected Result                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1  | Go to <b>Outbound &gt; Orders</b> and click <b>New Order</b> . Select <i>CV Gadget Store</i> as the customer. Add a line item for <i>Pro Laptop X</i> with a quantity of 15 and set the shipping address. | A new sales order is created with status <i>Draft</i> showing the line item, quantity and total price.                                   |
| 4.2  | Confirm the order. The system should allocate stock from available inventory (15 units) and change the order status to <i>Reserved</i> .                                                                  | The order status changes to <i>Reserved</i> . Inventory commitment shows 15 units reserved for the order.                                |
| 4.3  | Review the order details and ensure that the reserved quantity has been deducted from available stock while on-hand stock remains unchanged (since items are not yet picked).                             | Inventory counts reflect the reservation: available stock decreases by 15, reserved stock increases by 15, and total on-hand remains 20. |

### 5. Picking

| Step | Action                                                                                                                                          | Expected Result                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.1  | Navigate to <b>Outbound &gt; Picking</b> and start a picking session. Select the sales order created in step 4.                                 | The system generates one or more pick tasks for the order, specifying the locations (e.g., <i>MS-A1</i> and <i>MS-A2</i> ) and quantities to pick.                             |
| 5.2  | Assign the picking tasks to a worker. The worker scans each bin, retrieves the correct quantity and confirms the pick via the mobile interface. | As tasks are completed, the system updates the order line items as <i>Picked</i> . Inventory on-hand in the picked bins decreases by 15 units, and reserved stock is released. |

## 6. Packing and Shipping

| Step | Action                                                                                                                                                                                 | Expected Result                                                                                                                                                                           |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.1  | Proceed to <b>Outbound &gt; Packing</b> (if available). Consolidate the picked items into a package, assign a package ID or license plate number (LPN), and generate a shipping label. | A package record is created with the correct dimensions and weight. The order status transitions from <i>Picked</i> to <i>Packed</i> .                                                    |
| 6.2  | In <b>Outbound &gt; Shipments</b> , create a shipment record for the packaged order. Enter carrier details and booking information, then mark the shipment as <b>Shipped</b> .         | The order status updates to <i>Shipped</i> , and a tracking number is recorded. Inventory for <i>Pro Laptop X</i> decreases by 15 on hand and the order appears in the shipments history. |
| 6.3  | (Optional) Navigate to <b>Invoices</b> and generate an invoice for the order.                                                                                                          | An invoice is created for <i>CV Gadget Store</i> with the correct quantity and price.                                                                                                     |

## Post-conditions and Validation

After completing the end-to-end scenario:

1. **Inventory Counts** – Verify that *Pro Laptop X* on-hand stock in the warehouse is now **5 units** (20 initially received – 15 shipped), located in the appropriate bins. Reserved stock should be zero.
2. **Purchase Order Status** – The purchase order created in step 1 should be marked as *Received* or *Done* with all quantities received.
3. **Sales Order Status** – The sales order should show status *Shipped* (and *Invoiced* if invoicing is enabled) with all line items fulfilled.
4. **Dashboard Metrics** – Navigate back to the **Dashboard**. The **Total Stock Value** should reflect the new inventory valuation. **Fulfilment Rate** should show at least one order shipped, **Pending Orders** should be zero, and **Warehouse Capacity** should update based on remaining stock. The **Daily Sales Trend** graph will show a spike for the shipped order.
5. **Audit and Logs** – Review system logs for each operation (purchase order creation, receiving, putaway, picking, packing, shipping) to ensure that timestamps, users and quantities have been recorded correctly.

## Notes and Variations

- **Negative Scenarios** – Include tests where received quantities differ from ordered quantities, where putaway rules fail (e.g., destination location is full), and where picks are short to verify error handling and exception flows.

- **Multiple Products** – Extend the scenario by adding additional SKUs with different storage requirements (e.g., refrigerated goods) to test multiple putaway rules and picking zones.
- **Integrated Devices** – If mobile scanners or voice-picking devices are used, test integration by scanning barcodes/QR codes at each stage and verifying that data synchronises correctly with the WMS.
- **Carrier Integration** – For shipments, test integration with carriers (e.g., Lalamove) for rate shopping and booking if the system is configured for that functionality.

This end-to-end scenario will exercise the core inbound and outbound processes of the Labamu WMS and highlight any gaps before deploying the system to production.

## Custom Workflow Setup and Activation

Many warehouse processes depend on configurable workflows to orchestrate inbound, outbound and cross-docking steps. The updated WMS includes a **Workflow Templates** module and a drag-and-drop **builder** for designing custom flows (as shown in the user-supplied screenshots). The following steps outline how to create, configure and activate a custom workflow that can be triggered manually or by a system event (e.g., purchase order receipt).

### Pre-conditions

1. **Workflow Permissions** – Ensure the test user has permissions to create, edit and activate workflow templates. This may require the *Workflow Designer* or *Administrator* role in the RBAC configuration.
2. **Step Library** – The workflow builder should have the necessary step types available (e.g., *Receive/Inbound*, *Put-Away*, *Cross-Dock*, *QC Inspect*, *Wave Pick*, *Staging*, *Packing*, *Shipping* and *Condition (IF/ELSE)*).

### Workflow Setup Steps

| Step      | Action                                                                                                                                                                                                                                                                                                                                                 | Expected Result                                                                                                                                                 |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>W1</b> | Navigate to <b>Workflows &gt; Templates</b> from the left navigation and click <b>New Workflow</b> . Enter a descriptive name (e.g., <i>Custom E2E Workflow</i> ), version (optional), description and select a trigger. For an inbound flow, choose <b>PO_RECEIPT</b> ; for a manually started test, choose <b>MANUAL</b> . Save to open the builder. | A new draft workflow template is created and the builder canvas opens with an empty grid. The workflow appears in the templates list with status <i>Draft</i> . |
| <b>W2</b> | From the <b>Step Library</b> panel on the left, drag a <b>Receive/Inbound</b> block onto the canvas. Select it and check <i>Is Starting Step</i> in the properties panel so that this step initiates the workflow.                                                                                                                                     | The Receive step appears on the canvas with a blue border (indicating the start). The properties panel shows it is the starting step and lists available        |

| Step      | Action                                                                                                                                                                                                                                                                                                                                                     | Expected Result                                                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                                                                                                                                                                                                                                                                                                                            | transitions.                                                                                                                                                                                                                           |
| <b>W3</b> | Drag a <b>Condition (IF/ELSE)</b> block onto the canvas below the Receive step. In the properties panel, rename it (e.g., “Check Cross Dock?”) and set up two transitions: one to a <b>Cross-Dock</b> block and one to a <b>Put-Away</b> block. Use drag-and-drop arrows to connect the Receive step to the condition and the condition to the next steps. | The canvas displays arrows connecting <i>Receive</i> → <i>Check Cross Dock?</i> → <i>Cross-Dock</i> and <i>Receive</i> → <i>Check Cross Dock?</i> → <i>Put-Away</i> . The properties panel lists both transitions.                     |
| <b>W4</b> | Drag <b>Cross-Dock</b> and <b>Put-Away</b> blocks from the library onto the canvas. Optionally, drag additional steps such as <b>QC Inspect</b> , <b>Wave Pick</b> , <b>Packing</b> and <b>Shipping</b> to model a complete process. Connect each step sequentially or branch as needed.                                                                   | The workflow canvas shows a complete flow from receiving to shipping, with conditional branching for cross-dock versus regular put-away. Each step’s properties can be configured (e.g., set <i>Cross-Dock</i> as a destination step). |
| <b>W5</b> | Click <b>Validate</b> in the builder toolbar. Resolve any validation errors (e.g., missing transitions or unconnected steps) until the workflow passes validation.                                                                                                                                                                                         | The system indicates that the workflow is valid. No error messages remain.                                                                                                                                                             |
| <b>W6</b> | Click <b>Save Workflow</b> to persist the draft. If modifying an existing template, you may also click <b>Save as New Version</b> . Then click <b>Back</b> to return to the <b>Templates</b> list.                                                                                                                                                         | The workflow appears in the templates list with status <i>Draft</i> or with an incremented version number. The last updated timestamp reflects the save action.                                                                        |
| <b>W7</b> | On the templates list, locate the new workflow and click <b>Activate</b> . Confirm activation.                                                                                                                                                                                                                                                             | The workflow’s status changes from <i>Draft</i> to <i>Active</i> . The trigger (e.g., <b>PO_RECEIPT</b> or <b>MANUAL</b> ) is now live, and the workflow will be invoked when the specified event occurs or when manually started.     |

## Activation Verification

1. **Trigger Test** – If the workflow trigger is **PO\_RECEIPT**, create and receive another purchase order as described in the inbound scenario (Steps 1 and 2). Verify that the newly activated workflow executes: tasks in the **Workflow Monitor** should show the workflow starting with the *Receive* step and proceeding through the defined steps. If

the trigger is **MANUAL**, navigate to **Workflows > Monitor**, select the workflow, and click **Start** to run it manually.

2. **Cross-Dock Branch** – For a purchase order flagged for cross-docking, verify that the workflow follows the *Cross-Dock* branch and skips the put-away step. For regular receipts, ensure the flow proceeds to *Put-Away* and subsequent steps (QC, wave pick, etc.).
3. **Error Handling** – Test validation by deliberately leaving a step unconnected or missing a transition; confirm that the builder's *Validate* button catches the issue and that the workflow cannot be activated until it is resolved.

This custom workflow setup ensures that inbound and outbound processes can be tailored to specific operational needs, including cross-docking, inspection, staged picking and direct shipping. Activating the workflow makes it available for real-time execution and integration with the rest of the end-to-end test scenario.

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